

Course Code: EET 305

Course Name: SIGNALS AND SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- 1 A rectangular signal $x(t)$ is shown in Fig. 1. Sketch the following signals: i) $x(t-2)$ 3
 ii) $3x(t)$ iii) $x(t-3) + 3x(t)$

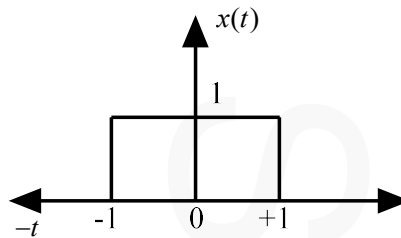


Fig. 1

- 2 A system is described by the input-output equation, $y(t) = x(t+1) + x(t^2)$. Determine 3
 whether the system is static, causal, time invariant, linear and stable.
- 3 State and prove the time-shifting property of Fourier series 3
- 4 Find the Fourier transform of $x(t) = \delta(t-2)$ 3
- 5 The signal flow graph of a system is shown in Fig. 2. Obtain the transfer function, 3
 $\frac{C(s)}{R(s)}$

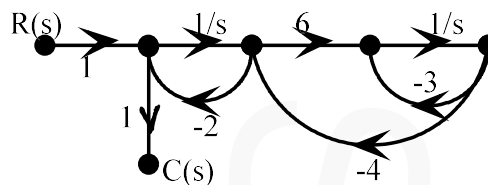


Fig. 2

- 6 By means of Routh's criteria, determine the stability of the system represented by the 3
 characteristic equation, $s^4 + 2s^3 + 8s^2 + 4s + 3 = 0$
- 7 Find the Nyquist rate and Nyquist width of the signal, $x(t) = (\sin 200\pi t)^2$ 3

- 8 Derive the transfer function of a ZOH circuit. 3
- 9 State and prove the time shifting property of discrete Fourier transform. 3
- 10 Find the discrete time Fourier series of $x[n] = \sin 0.2\pi n$. 3

PART B

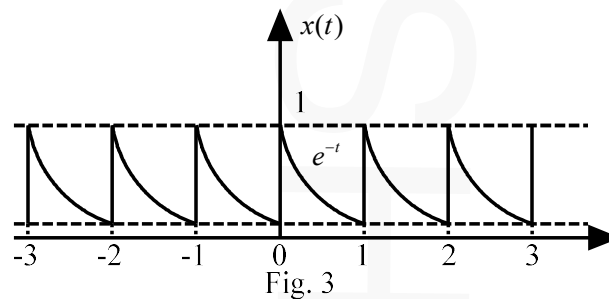
(Answer one full question from each module, each question carries 14 marks)

Module -1

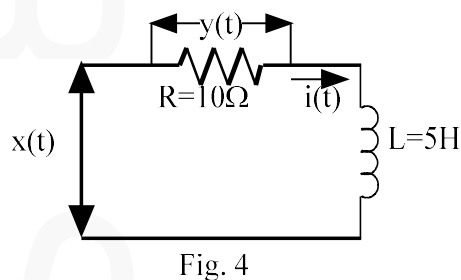
- 11 a) Explain random variables and random processes 7
- b) Find the output of a system with impulse response $h(t) = (2 - e^{-2t})u(t)$ and the input signal, $x(t) = e^{-3t}u(t)$ 7
- 12 a) Find the odd and even components of the signal, $x(t) = \cos t + \sin t + \cos t \sin t$ 7
- b) Differentiate energy and power signals. Determine the energy and power of the signal, $x(t) = 5 \cos(10t + \phi) + 10 \sin(5t + \phi)$ 7

Module -2

- 13 a) Find the exponential Fourier series coefficients for the signal shown in Fig. 3 9



- b) State and prove the time integration property of Fourier transforms 5
- 14 a) Find the unit step response of the circuit shown in Fig.4 7



- b) Determine the output response of the system whose impulse response $h(t) = e^{-2t}u(t)$, for a unit step input. 7

Module -3

- 15 a) Reduce the block diagram shown in Fig.5 and obtain the transfer function $\frac{C(s)}{R(s)}$ 10

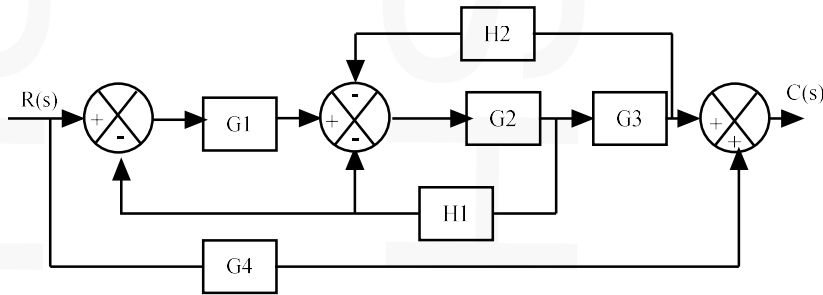


Fig. 5

- b) State Mason's gain formula and explain the terms involved 4
- 16 a) State and explain the Hurwitz criterion for analysing the stability of LTI systems 4
- b) Determine the range of values of K that stabilises the closed loop system shown in Fig. 6. 10

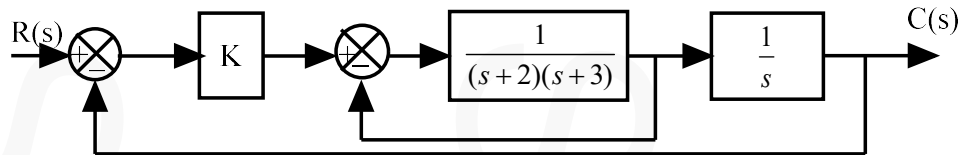


Fig. 6

Module -4

- 17 a) Find the z- transform of $x[n] = a^n \cos \omega_0 n u[n]$ 5
- b) A linear, time- invariant system has the impulse response, $h(n) = [u(n) - u(n-6)]$. 9
The system is excited by $x(n) = [u(n-1) - u(n-5)]$. Determine the output of the system.
- 18 a) The input to a causal LTI system is $x[n] = u[-n-1] + \left(\frac{1}{2}\right)^n u[n]$. The z- transform of 10
the output of the system is $Y[z] = \frac{-\frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)(1 + z^{-1})}$. Determine $H[z]$, the z-
transform of the impulse response and also determine the output $y[n]$
- b) Explain aliasing effect in sampled data systems. 4

Module -5

- 19 a) Solve the difference equation, $y[n] + 6y[n-1] + 8y[n-2] = 5x[n-1] + x[n-2]$. 8

The initial conditions are $y[-1] = 1$; and $y[-2] = 2$. The input $x[n] = u[n]$.

- b) Find the discrete Fourier series coefficients and Fourier series for the function, 6

$$x[n] = \sin^2\left(\frac{\pi}{6}n\right)$$

- 20 a) Obtain the parallel form realization of the system function, $H[z] = \frac{z^2 + 4z + 10}{(z+2)(z+4)}$ 7

- b) Check the stability of the system whose characteristic equation is given by , 7

$$z^4 - 1.7z^3 + 1.04z^2 - 0.268z + 0.024 = 0, \text{ using Jury's stability criterion.}$$
